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# MDF-FND: A dynamic fusion model for multimodal fake news detection

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ABSTRACT

Fake news detection has received increasing attention from researchers in recent years, especially in the area of multimodal fake news detection involving both text and images. However, many previous studies have simply fed the semantic features of both text and image modalities into a binary classifier after applying basic concatenation or attention mechanisms, where these features often contain a significant amount of inherent noise. This, in turn, leads to both intra- and inter-modal uncertainty. In addition, while methods based on simple concatenation of the two modalities have achieved notable results, they often ignore the drawback of applying fixed weights across modalities, which causes some high-impact features to be ignored. To address these issues, we propose a novel semantic-level **m**ultimodal **d**ynamic **f**usion framework for **f**ake **n**ews **d**etection (**MDF-FND**). To the best of our knowledge, this is the first attempt to develop a dynamic fusion framework for semantic-level multimodal fake news detection. Specifically, our model consists of two main components: (1) the **U**ncertainty **E**stimation **M**odule (**UEM**), which is an uncertainty modeling module that uses a multi-head attention mechanism to model intra-modal uncertainty, and (2) the **D**ynamic **F**usion **N**etwork, which is based on Dempster–Shafer evidence theory (**DFN**) and is designed to dynamically integrate the weights of both text and image modalities. To further enhance the dynamic fusion framework, a graph attention network is employed for inter-modal uncertainty modeling before DFN. Extensive experiments have demonstrated the effectiveness of our model across three datasets, with a performance improvement of up to 4% on the Twitter dataset, achieving state-of-the-art performance. We also conducted a systematic ablation study to gain insights into our motivation and architectural design. Our model is publicly available at https://github.com/CoisiniStar/MDF-FND.

## 1. Introduction

With the rapid development of the Internet, especially the rise of multimedia, multimodal posts both text and images have become increasingly common on social media platforms, such as Twitter and Weibo. An increasing number of people share what they see and hear on these platforms, providing timely updates on social events happening worldwide. However, of these posts also contain fake news. In recent decades, the frequent occurrence of major emergencies has contributed to the proliferation of fake news, which has had a significant impact on society, often leading to widespread panic. For example, during the 2016 U.S. presidential election [1], the proliferation of fake news on social media seriously affected voters' intentions, undermining both the fairness of the election process and its outcome. Moreover, during the COVID-19 pandemic [2], social media was flooded with intentionally fabricated misinformation, including claims that saltwater, tea, and vinegar could eliminate COVID-19.

With the emergence of multimodal news containing both text and images, detecting fake news has become increasingly challenging, prompting researchers to focus on multimodal fake news detection. Vinyals et al. [3] were the first to propose multimodal fusion, generating natural sentences to describe images. Jin et al. [4] built an end-to-end network for fake news detection using a recurrent neural network (RNN) with a local attention mechanism, which combined text, images, and social context features. To improve the detection of fake news, Qu et al. [5] applied a quantum multimodal fusion strategy, further improving detection performance. While these previous studies achieved promising results, they still face the following challenges:

1. Multimodal data come from heterogeneous sources [6], often of low quality and containing noise [7,8]. For instance, in the two examples of multimodal posts shown in Fig. 1, the news texts in the first example are non-normalized, which can negatively impact classification results. In the second example, the attached

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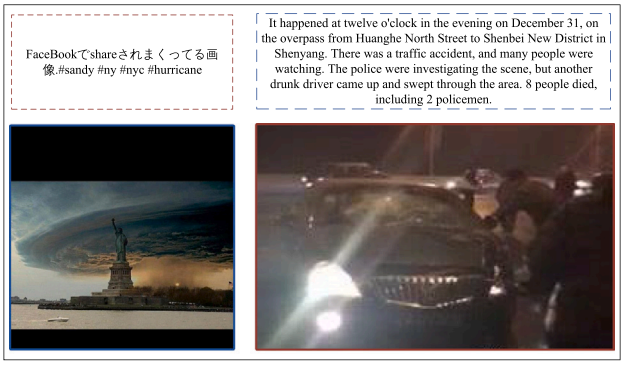



**Fig. 1.** Examples of low-quality multimodal posts. The first image illustrates irregular news tweets, while the second image displays an attached image with low resolution.

image has low resolution, causing the extracted features to include non-excludable noise. However, as shown in Fig. 2(a), existing research [9–12] rarely accounts for data uncertainty caused by inherent noise in each modality and typically employs a simple static fusion strategy, leading to noise accumulation between modalities and reducing the model's generalization performance.

2. Some traditional methods [11–13] perform simple fusion by combining feature vectors obtained from text and visual encoders and then feeding the fused features into a classifier. Each modality maintains fixed weights, making it unable to assign higher confidence levels to features with high impact factors [14]. In addition, this approach implicitly assumes that the quality of all samples across different modalities is stable. However, in reality, the quality of different modalities for different samples can vary significantly, as shown by the two examples in Fig. 1.

Multimodal features are frequently affected by the impact of noise in high-dimensional data, which can adversely impact the final fusion results [15]. To better integrate posts and images from heterogeneous data sources, inspired by the concept of trusted multi-view classification [16], we propose a framework for multimodal dynamic fusion applied to fake news detection. The main motivation of this work is to develop a dynamic decision-making mechanism for multimodal news samples in which the quality varies significantly, based on the assessment of their uncertainty. Specifically, to reduce the uncertainty resulting from noise accumulation, it first uses an **U**ncertainty **E**stimation **M**odule (**UEM**) to model the uncertainty within each modality. The robust feature representations obtained are input into the **D**ynamic **F**usion **N**etwork (**DFN**) to dynamically assess the quality of different modalities, allowing for the dynamic fusion of features, as shown in Fig. 2(b). When uncertainty is high across all modalities, a multimodal heterogeneous graph structure is relied upon to model inter-modal uncertainty and generate the final decision value. Conversely, when uncertainty is lower, we rely on the decision outcomes from the **D**ynamic **F**usion **N**etwork (**DFN**), which is based on Dempster–Shafer (D–S) evidence theory. Experiments conducted using the MDF-FND framework on three public datasets — Twitter, Weibo, and PolitiFact — demonstrate that our model delivers exceptional performance in multimodal fake news detection.

Recent advances in large language models (LLMs) have led to outstanding performance in various natural language processing (NLP) tasks; however, they still face significant challenges in multimodal fake news detection. For example, as noted by [17], although complex LLMs such as GPT3.5 can provide valuable multi-perspective justifications for exposing fake news, they are still less effective than smaller language models (SLMs) such as fine-tuned BERT in these tasks. Therefore, we focused on fine-tuning a small model specifically for multimodal fake news detection.

The main contributions of this study are as follows:

- We developed an uncertainty modeling module utilizing multi-head attention to effectively extract robust feature representations from noisy multimodal news content.
- We designed a multimodal DFN based on D–S evidence theory, which dynamically balances the contributions of both modalities in the final classification, effectively addressing the limitations of weight fixation in static fusion strategies.
- Our experimental results on three real-world multimodal benchmark datasets demonstrate the effectiveness and superiority of our model.

The remainder of this paper is organized as follows. Section 2 review related work on fake news detection and relevant methods. Section 3 formalizes the MDF-FND framework and provides a detailed explanation of the design for each layer. The experimental setup and results are presented in Section 4. The limitations of this study are discussed in Section 5. Finally, Section 6 summarizes our work and outlines future research directions.

## 2. Related work

In this section, the distinction between multimodal and multiview data is shown, followed by briefly review existing research on multimodal fake news detection. Relevant studies on multimodal dynamic fusion strategies and D–S evidence theory are also the most important background, as these techniques are closely related to our approach.

### 2.1. Feature selection in multi-view learning and multi-modal learning

Multi-view learning refers to the technique of using multiple different representations or descriptions of the same data object, typically originating from the same type of sensor or modality. Each "view" reflects a distinct perspective of the data, potentially observed from different times, locations, or conditions. Recently, some researchers have focused on the influence of feature selection in multiview learning. For example, Han et al. [18] explored the relationship between features, views, and labels, and proposed a new method that integrates feature correlation and redundant representation. Specifically, for multiviews, particularly in multi-label data, the feature correlation component uses label information coefficients to assess the information gain of each label, while the feature redundancy component uses view redundancy factors to address feature redundancy both within and across views. This approach effectively strikes a balance between acquiring label information and reducing feature redundancy. Hao et al. [19] examined the inconsistencies present in each view, including noise and specific elements, which may affect the quality of the mapping between labels and feature representations. Moreover, neglecting certain aspects could lead to suboptimal outcomes, as each label should have its own distinctive features. To address this, they proposed a unified loss function, which is a completely partitioned structure for the observed labels as hybrid labels, i.e., common labels, view-specific labels, and noise labels. The view-specific labels are further subdivided into several specific labels for each view. This method simultaneously takes into account the consistency and complementarity of different views. By examining the feature weights of the hybrid labels, the mapping relationship between labels and features can be established sequentially based on their attributes. Additionally, their work investigates the correlations between hybrid labels and incorporates these relationships into the loss function.

Multimodal learning involves different modalities (perception types or data forms) of the same data object, typically drawing from various sensory sources. Each modality may encompass various types of sensory data, such as vision, audio, text, sensor data, and more. Typically, multimodal data suffers from the curse of dimensionality,

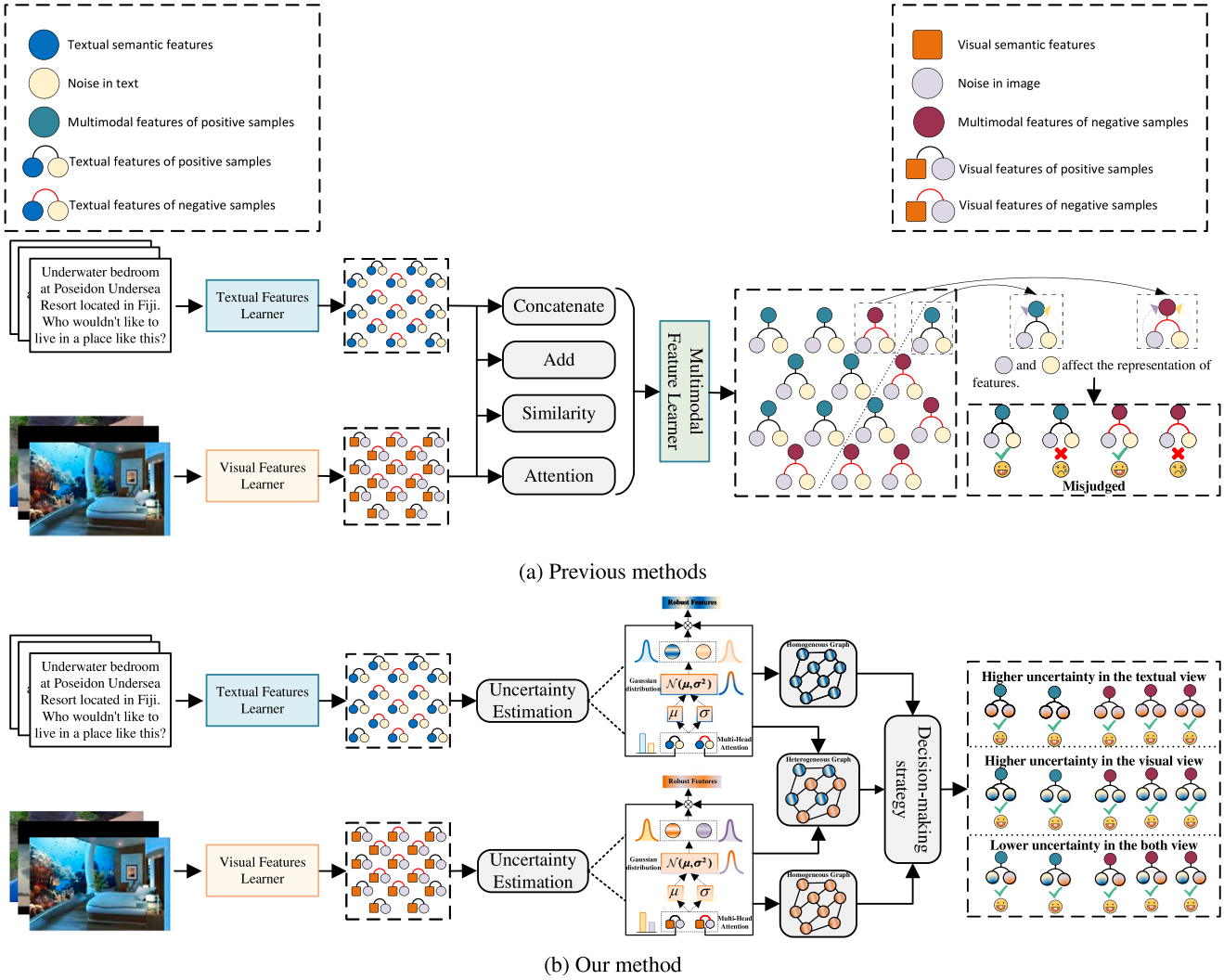



**Fig. 2.** Comparison of the fusion and reasoning processes. (a) Previous methods have employed basic cross-modal interaction operations, resulting in a higher likelihood of misclassification due to the final fused features. (b) Our method overcomes the limitations of previous static fusion approaches by incorporating uncertainty modeling and dynamic feature integration. Moreover, it amplifies the feature-level distinction between real and fake news in the presence of noise, thereby improving the robustness of classification.

making feature selection especially important for multimodal fusion. For example, Zhang et al. [20] observed that both the candidate and selected features contribute valuable classification information when designing the feature correlation term. As a result, they extracted relevant classification information from the joint mutual information and proposed a new feature correlation term called Conditional Weight Joint Relevance (CWJR). Based on CWJR, they proposed a feature selection method named CWJR-FS, which incorporates the concept of conditional weight joint relevance. Zhang et al. [21] examined the influence of selected features on feature correlation, as well as the effect of label correlation on feature correlation. They designed a new feature correlation term called "Dual Conditional Correlation" (DCR), which employs two conditional mutual information terms to account for the correlation between selected features and labels. Gao et al. [22] proposed an information-theory-based feature selection method: similarity retention and gaze-based decision. This method first leverages the similarity between samples to generate transformed data, mitigating outliers or erroneous features in the original data. Additionally, the new transformed data improves the discriminability of features, which helps enhance the quality of the data.

Multiview focuses on multiple angles or representations of the same type of data. In contrast, multimodal focuses on different types of data sources (such as images, text, audio, etc.). Additionally, the data sources in view-based methods are usually of the same type (e.g., images from different perspectives), whereas in multimodal methods, the data sources come from different modes of perception (e.g., text and image, video and audio, etc.). Thus, multi-view feature selection emphasizes different representations of the same dataset, while multimodal feature selection focuses on integrating heterogeneous data sources. In this paper, we focus on the multimodal perspective of fake news detection.

### 2.2. Multimodal fake news detection

As multimodal data continue to evolve, several researchers have become increasingly enthusiastic about processing tasks that involve both text and images. Antol et al. [13] proposed a visual question-and-answer task at learning feature representations from multiple aspects. This work promotes cross-disciplinary research in image understanding and NLP, laying a foundation for multimodal fake news detection. The event adversarial neural network (EANN) model introduced by Wang et al. [9] employs an event learner to capture feature representations from both text and images in news articles; however, the inclusion of auxiliary features increases detection costs. MVAE [11] extracts intra-modal information from two unimodal modalities, text and images, utilizing a multimodal variant encoder for simple fusion.

**Table 1**
A comparative analysis of the feature fusion mechanisms used in existing SOTA techniques.

| Model | Feature fusion mechanism | Dataset ♠ | Accuracy | Precision ♣ | Recall ♣ | F1 score ♣ |
|---|---|---|---|---|---|---|
| SpotFake [10] | Concatenation | Twitter [23] | 77.0% | 67.8% | 86.6% | 75.3% |
| CMMTN [24] | Concatenation | Twitter [23] | 90.3% | 89.8% | 89.9% | 89.7% |
| MPFN [25] | Concatenation | Twitter [23] | 83.3% | 82.7% | 82.1% | 81.0% |
| EANN [9] | Concatenation | Twitter [23] | 71.5% | 82.2% | 63.8% | 71.9% |
| BDANN [26] | Concatenation | Weibo [4] | 81.4% | 82.0% | 81.0% | 81.5% |
| SAFE [12] | Concatenation | PolitiFact [27] | 87.2% | 90.4% | 83.7% | 86.7% |
| KATMF [28] | Concatenation | COVID-19[28] | 92.5% | 92.5% | 92.3% | 92.4% |
| MCNN [29] | Similarity | Twitter [23] | 82.3% | 82.2% | 82.4% | 82.2% |
| MCAN [30] | Attention | Weibo [4] | 89.9% | 89.8% | 89.9% | 89.9% |
| COOLANT [31] | Attention | Weibo [4] | 92.3% | 92.3% | 92.2% | 92.2% |
| CAFE [32] | Interaction matrix | Weibo [4] | 84.0% | 84.0% | 84.0% | 83.9% |
| MVAE [11] | Sampling-based reconstruction | Twitter [23] | 74.5% | 74.5% | 74.8% | 74.4% |

♣ The metric is based on the average across the positive and negative samples.
♠ Most models are evaluated on multiple datasets, but we have selected only the best-performing one.

SpotFake [10] uses a pre-trained language model to extract features, leveraging large-scale textual and visual pre-trained models to create a multimodal representation vector by simply concatenating features from each unimodal state, which is then fed into a binary classifier for fake news detection. Similarly the SAFE [12] model employs simple fusion to detect fake news through a similarity-based approach. Wu et al. [30] proposed the multimodal common attention network (MCAN) for fake news detection, which learns the inter-dependencies between multimodal features and achieves commendable results. Wang et al. [33] proposed a fake news detection framework called MetaFEND, which is effective in detecting breaking news because it learns from a small number of verified posts. However, this method increases the cost of detection. Song et al. [28] added a graph convolutional network (GCN) to their framework, called KATMF, using textual information fused with knowledge concepts and visual information for multimodal modal semantic representation. In addition, other researchers, such Xue et al. [29] and Fung et al. [34], have explored the consistency of multimodal data to help detect fake news. Wang et al. [31] developed the COOLANT framework using powerful cross-modal contrastive learning methods. This framework aims to achieve more accurate image–text alignment. To further capture the fine-grained alignment between the visual and linguistic elements, they employed auxiliary tasks to mitigate the loss associated with negative samples during the contrastive process. Zhang et al. [26] introduced an end-to-end model named BDANN, which is a BERT-based domain-adaptive neural network designed for effective detection of multimodal fake news. MPFN [25] can more accurately identify fake news by integrating multiple information sources, such as text and images. Specifically, this approach first extracts features from each information source and then combines these features using a progressive fusion strategy to obtain the final detection results. Compared to methods relying on a single information source, this approach captures the complex characteristics of fake news more effectively, thereby improving detection accuracy. Wang et al. [35] proposed an instance-guided multimodal graph fusion method that detects fake news by modeling both internal and external relationships between images and text. This method uses multimodal information to dynamically fuse the relationships between different modalities, enhancing the identification of fake news. Singhai et al. [36] introduced a multimodal fake news detection architecture that effectively identifies and suppresses information from weak modalities while extracting relevant information from strong modalities. CAFE [32] captures ambiguous relationships between different modalities, thereby enhancing the accuracy of fake news detection. Existing multimodal FND approaches typically directly fuse unimodal features into the multimodal representation before detecting the authenticity of news samples. However, this approach overlooks the authenticity neutralization effect across different modalities. Therefore, Wang et al. [37] proposed a novel detection framework named MINER-UVS, which utilizes unimodal authenticity signals in multimodal fake news detection. Wang et al. [24] observed that users on media platforms are likely to label only a small number of posts from categories they are interested in. As a result, they introduced positive unlabeled learning to the domain of fake news detection and proposed a curriculum-based multimodal masking transformer network named CMMTN for detecting positive unlabeled fake news.

The feature fusion mechanisms of existing SOTA techniques are presented in Table 1. As observed, in the existing detection frameworks, the multimodal features are obtained by performing concatenation on the learned unimodal features, where each modality contributes equally to the final multimodal feature. This fusion method is referred to as static fusion. As illustrated in Fig. 2(a), traditional static fusion approaches fail to account for the modality noise generated during unimodal feature learning. These methods simply concatenate each modality with fixed weights, disregarding the confusion level inherent in each modality's data. Although such methods have improved fake news detection performance to some extent, the features produced by the multimodal feature learner are often cluttered with noise interference from individual modalities. This ultimately affects the performance of the final detector. To address this, we propose a dynamic fusion mechanism based on D–S theory, assigning a confidence score to each modality and dynamically adjusts the modality weights accordingly.

### 2.3. Multimodal dynamic fusion

Multimodal dynamic fusion allows the model to take full advantage of the complementary features of different modalities, which in turn improves robustness and generalization performance. To better analyze human multimodal languages, Han et al. [38] proposed a new trustworthy multimodal classification algorithm that dynamically evaluates both the feature and modality levels of information for different samples, thus credibly integrating multiple modalities. Based on theoretical analyses, Zhang et al. [14] further revealed that the generalization ability of dynamic fusion is closely linked to the performance of uncertainty estimation. They proposed generalized dynamic multimodal fusion framework for low-quality data, which uses the generalization error derived from multimodal fusion to dynamically update each unimodal predictor. This approach enables the multimodal decision-making process to rely more heavily on higher-quality modalities while mitigating the influence of unreliable ones by dynamically determining the fusion weights for each modality.

Inspired by these studies, we propose a dynamic fusion mechanism for the fake news detection task that effectively balance the weights of each modality.

### 2.4. Graph-based multi-modal learning

In areas where multi-modal techniques are well developed, there are many graph-based methods have been proposed to capture cross-modal information. Yin et al. [39] propose a novel approach where the

sentence-image pair is treated as a unified graph, enabling the effective capture of diverse semantic relationships among multimodal semantic units. Liu et al. [40] proposed a methodology called TMac, a temporal multimodal graph learning method designed for Acoustic event Classification, which can effectively capture the dynamic information in intra-modal and inter-modal. Liu et al. [41] proposed a self-supervised method for temporal graph learning, called S2T, which can simultaneously extract temporal and structural information, thereby learning more informative node representations. This method not only models temporal information using historical neighbor sequences but also considers structural knowledge at both local and global levels. At the local level, it generates structural strength by aggregating the features of higher-order neighbor sequences. At the global level, it generates a global representation based on all nodes to adjust the structural strength according to the activity status of different nodes.

Thus, GNNs excel in multimodal data fusion. Consequently, a GAT mechanism is adopted to fully exploit the complementary information between modalities.

### 2.5. D–S evidence theory

D–S evidence theory, also known as confidence function theory, is a complete theoretical architecture that integrates multi-source information and is widely used in multimodal fusion tasks. D–S evidence theory was first introduced by [42]. Subsequently, [43] introduced mathematical evidence theory and applied it to multi-source information fusion. Currently, D–S evidence theory has a wide range of applications in the field of fusion decision-making [44]. The D–S evidence theory framework facilitates the development of Bayesian theory and simultaneously relaxes the limitations of Bayesian theoretical reasoning methods [45]. It aims to compute and update the confidence of a proposition by integrating evidence from multiple sources, serving as a mathematical framework for addressing uncertainty. However, traditional synthesis rules cannot be directly applied within the MDF-FND framework ; thus, we will elaborate on our algorithm in Section 3.3.

In our approach, we input robust feature representations generated by the uncertainty modeling module, which employs multi-head attention, into the fusion network to learn dynamic inter-modal weight modeling. Ultimately, D–S evidence theory is used to achieve the dynamic fusion of textual and visual modalities, establishing a comprehensive framework for multimodal fake news detection.

## 3. Methodology

In this section, we introduce a dynamic fusion framework for fake news detection that leverages D–S evidence theory and GAT. Section 3.1 provides an overview of the proposed MDF-FND. In Section 3.2, we detail the Uncertainty UEM, which utilizes an attention mechanism for effective feature extraction. Finally, Section 3.3 introduces the dynamic fusion module that integrates GATs with D–S evidence theory to enhance the fusion of multimodal data.

### 3.1. Overview

The architecture of MDF-FND is depicted in Fig. 3. To model noisy multimodal news, unimodal semantic features generated by pre-trained BERT [46] and Resnet-50 [47] are first fed into the UEM module, which uses a multi-head attention mechanism to map semantic features into a latent Gaussian subspace for unimodal uncertainty modeling. Traditional point embeddings are represented in this subspace by the mean $\mu$ and variance $\sigma$. A graph attention network, where the number of layers depends on the input feature dimension, is used to model the inter-modal uncertainty. As shown in Fig. 2, traditional fusion mechanisms tend to combine the inherent noise from each modality, resulting in noisy multimodal features. This data confusion can negatively affect model's final decisions. To address this issue, we introduce

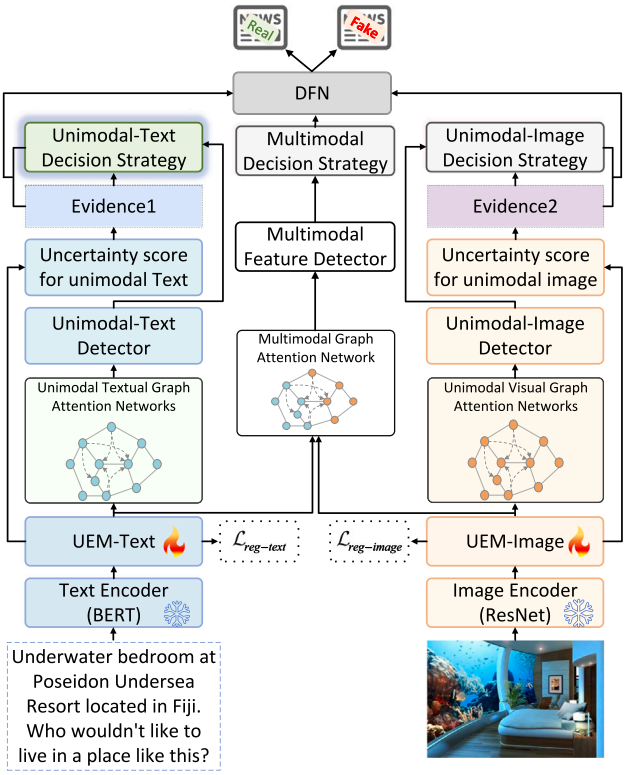



**Fig. 3.** Architecture of the MDF-FND framework. The MDF-FND comprises the UEM, which employs an attention mechanism and a GAT, along with the DFN incorporating D–S evidence theory and a fake news detector. Given a noisy post and its associated image from a social platform, the UEM maps the post and image into their respective latent subspaces, performing uncertainty modeling within each modality. Regularization terms $\mathcal{L}reg_{text}$ and $\mathcal{L}reg_{image}$ are applied to maintain the level of uncertainty in the distributions above a specified threshold $\eta$. The features produced by the UEM are then processed by the GAT to model inter-modal uncertainty, while the DFN facilitates dynamic weight adjustment between the modalities. Finally, different decision rules are applied based on the confidence levels of each modality.

a dynamic fusion approach. Specifically, we calculate each modality's data confusion level based on the robust features generated by UEM, creating an uncertainty score for each modality. These scores are then fed into the DFN, which selects the optimal decision rule based on evidence derived from both the visual and textual perspectives.

### 3.2. Modeling unimodal uncertainty using the multi-head attention mechanism

Most social media posts exist in multimodal form (containing text and their accompanying images). First, a pre-trained language model (PLM) is used to learn the token-level embedding representations of posts for extracting linguistic features. Specifically, let $T = ([CLS], t_1, t_2, \ldots, t_n, [SEP])$ be a sequence of post texts with $n$ tokens, and let $e^t = (e_1, \ldots, e_n)$ be an embedded representation of the context token level generated by the pre-trained BERT [46], where $e^t \in R^{n\times d}$ and each token $e_i \in \mathcal{R}^d$ is a $d$-dimensional vector. Similarly, we follow many previous studies [9,11,48] by using the popular pre-trained backbone network (i.e., pre-trained Resnet-50) to extract region-of-interest (RoI) pooling features from post attachments to obtain fine-grained object-aware representations. Formally, $e^v = \{e_1^v, e_2^v, \ldots, e_l^v\}, e^v \in \mathbb{R}^{l\times d_v}$, where $e_i^v \in \mathbb{R}^{d_v}$ denotes the $i$th ROI feature, and $i$ is the number of extracted ROI features. However, images accompanying user tweets are often of low resolution, and much of the textual content is non-standardized, resulting in significant noise in multimodal news data. Previous research [9–12] has not considered the incidental uncertainty caused by data noise. To effectively model the noisy posts consisting

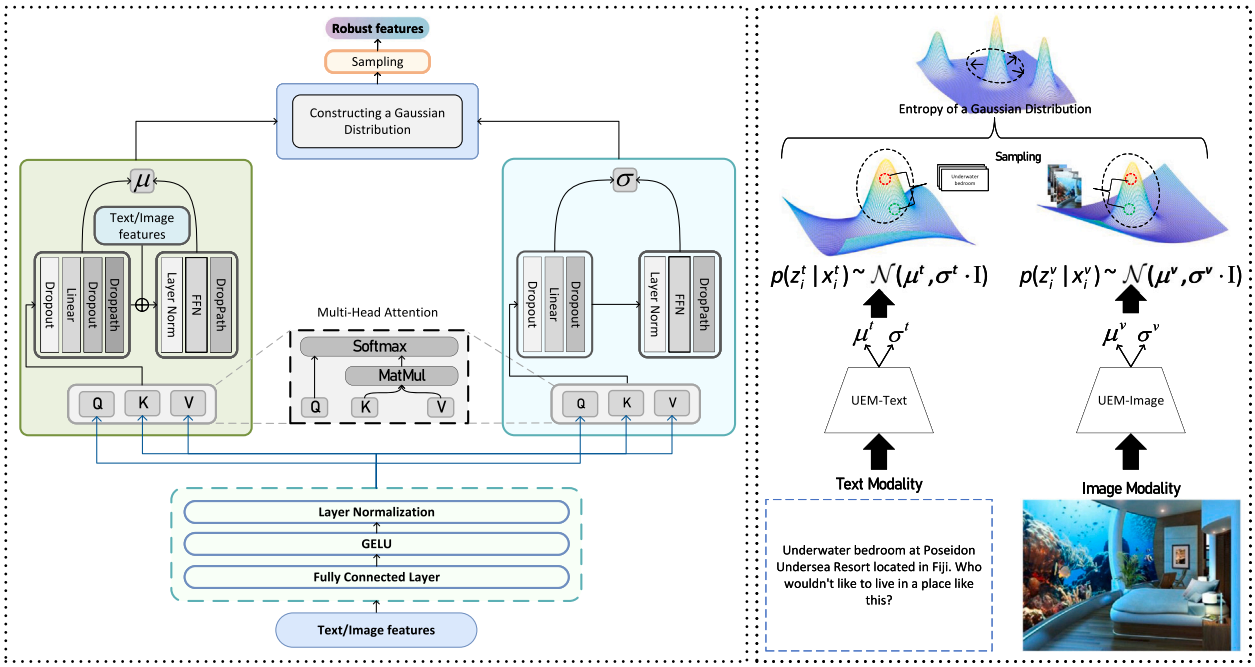



**Fig. 4.** Architecture of the UEM. The UEM utilizes a multi-head attention mechanism to represent the features of each modality as a Gaussian distribution characterized by a mean $\mu$ and variance $\sigma$. Samples are drawn from this distribution to generate robust feature representations, enabling effective uncertainty modeling in multimodal data.

of images and text, the UEM based on the multi-head attention mechanism is utilized to conduct uncertainty modeling for both modalities. Specifically, we use a multivariate Gaussian distribution to model the extracted features of the text and image, representing them as Gaussian distributions. The embedding representation of each modality is reconstructed using samples drawn from these distributions. We assume that the noise for each modality follows a Gaussian distribution. The UEM module computes the mean vector $\mu$ and variance vector $\sigma$ for the input features of each modality The mean vector indicates the location of the Gaussian distribution in the probability space, while the variance vector represents the spread across each dimension; its architecture is illustrated in Fig. 4. In the multi-head attention operation, the input representation $H \in \mathbb{R}^{T\times D}$ is divided into $k$ heads, where $T$ denotes the unimodal feature length and $D$ indicates the size of the hidden layer. Two channels are used in the UEM module to represent the mean $\mu$ and variance $\sigma$ of the generated Gaussian distribution. Because we use a multi-head attention mechanism, the input representation $H^{(i)} \in R^{T\times D/2k}$ is projected onto the query vectors $Q^{(i)}$, $K^{(i)}$, $V^{(i)}$, where $i$ denotes the number of heads. Formally, the multi-head attention used to generate the mean $\mu$ satisfied by unimodal textual noise can be expressed as

$$[Q^i_\mu(T), K^i_\mu(T), V^i_\mu(T)] = [W_q(T), W_k(T), W_v(T)]^T \times H^{(i)}_\mu(T) \tag{1}$$

The multi-head attention used to generate the mean $\mu$ satisfied by unimodal image noise can be expressed as

$$[Q^i_\mu(V), K^i_\mu(V), V^i_\mu(V)] = [W_q(T), W_k(V), W_v(V)]^T \times H^{(i)}_\mu(V) \tag{2}$$

We specify that $d_k = D/(2k)$. In the above equation, $W_q(T) \in \mathbb{R}^{d_k\times 3d_k}$, $W_k(T) \in \mathbb{R}^{d_k\times 3d_k}$ and $W_v(T) \in \mathbb{R}^{d_k\times 3d_k}$ are the weighting matrices used to project the features onto each header subspace with a weight matrix. $H^{(i)}_\mu(T)$ and $H^{(i)}_\mu(V)$ imply unimodal input features at the $i$th attention head for the text and image, respectively. $Q^i_\mu(T)$, $K^i_\mu(T)$, and $V^i_\mu(T)$ are the query, key, and value vectors formed based on textual features, respectively. Similarly, $Q^i_\mu(V)$, $K^i_\mu(V)$, and $V^i_\mu(V)$ are the query, key, and value vectors formed based on image features, respectively. The attention vector for variance $\sigma$ follows a similar formulation as that of the mean and will be described in the context of the mean. Furthermore, we adhere to the methodology outlined in [49], where a scaling strategy for the dot product attention is employed for each attention head. For example, Eq. (3) indicates the output of the $i$th head used to generate the unimodal text noise mean $\mu$:

$$H^{(i)} = Action\left(\frac{Q^{(i)}_\mu(T)K^{(i)}_\mu(T)}{\sqrt{d_k}}\right) \cdot V^{(i)}_\mu(T) \tag{3}$$

where $Action$ denotes the activation and normalization functions. $Q^i_\mu(T)$, $K^i_\mu(T)$, and $V^i_\mu(T)$ indicate the query, key, and value vectors for generating the mean value of the unimodal textual noise $\mu$ formed by the $i$th attention head, respectively. Finally, the vectors from the plurality of attention heads representing the mean value $\mu$ of the noise present in the unimodal text modality are concatenated to form a robust representation of the unimodal text:

$$MH^{Total}_\mu(T) = W_{out}(T) \cdot concat_{i\in[k]}[H^{(i)}_\mu(T)] \tag{4}$$

where $concat$ indicates that each attention head $H^{(i)}$ from the mean satisfied by the modal noise used to generate a single post or image is subjected to a concatenation operation along a certain dimension. A learnable weight matrix $W_{out}$ is introduced into the above equation to linearly transform the features after joining each attention head. For the final unimodal characterization, the sampling reparameterization strategy is used. However, since the sampling approach is a non-differentiable operation, it presents challenges in suppressing gradient backpropagation. To address this issue, we draw inspiration from the work of Wei et al. [48] and introduce reparameterization into our architecture. This method lies in decoupling the computational operations from the sampling process within the network. This decoupling enables the entire network to be optimized end-to-end using gradient descent. Sampling operations, being non-differentiable by nature, present a challenge for optimization through backpropagation. However, by decoupling the two processes, we can separate computation from sampling, which allows the network to remain differentiable even when non-smooth sampling operations are present. Specifically, we sample a random variable from a standard normal distribution rather than performing the sampling operation directly within $\mathcal{N}(\mu, \sigma^2)$:

$$z = \mu + \varepsilon\sigma \tag{5}$$

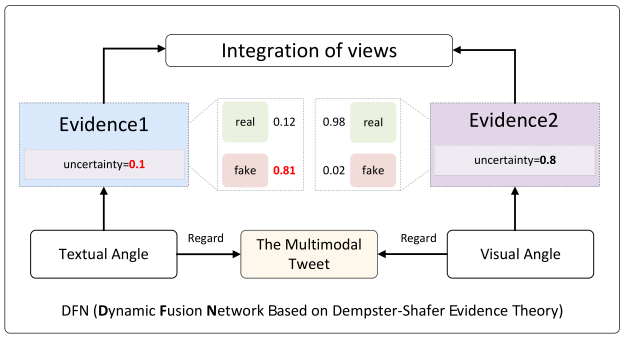



**Fig. 5.** Architecture of the DFN. The DFN integrates evidence from both textual and visual perspectives to make decisions on multimodal posts, utilizing D–S evidence theory. This evidence is derived from the unimodal confidence scores defined in Eqs. (10) and (11). The confidence score is inversely related to uncertainty. Based on the confidence scores of each modality, the DFN selects one of the following strategies for the final decision: multimodal decision strategy, unimodal-text decision strategy, or unimodal-image decision strategy. It then outputs the corresponding decision value.

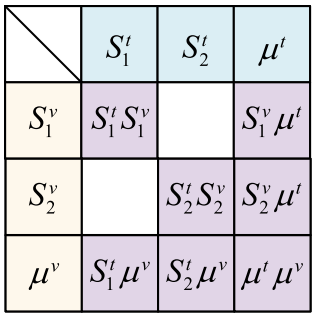



**Fig. 6.** Diagram illustrating the impact of binary classification utilizing D–S evidence theory. The light-blue region denotes the confidence score derived from the single text modality, and the light-yellow region represents the confidence score derived from the single image modality. The white region indicates the conflict area between the two decisions. The light-purple area highlights the region where the decision strategies of the two modalities align compatibly.

After Eq. (5), we output $z$ based on the predictive distribution derived from the UEM. This approach effectively separate the computation of the mean and standard deviation from the sampling operation, allowing for a decoupling of the original parameters, which transforms them into trainable ones.

Following the UEM module, we obtain a more robust feature representation for each modality, as it captures the distribution characterizing the noise inherent in each modality, in contrast to the traditional point embedding approach. This robust feature representation is particularly advantageous for detecting multimodal fake news, as discussed in detail in the subsequent ablation experiments.

### 3.3. DFN module based on GAN and D–S evidence theory

The simple concatenation mechanism has been employed in several previous multimodal fake news detection tasks [10–12], where fixed fusion weights are applied to the text and image unimodalities. However, certain posts may exhibit high uncertainty due to various factors. To address this issue, the DFN module is designed to incorporate the confidence of each modality within its internal structure. The architecture of the DFN is illustrated in Fig. 5. Intuitively, evidence 1 from the textual perspective exhibits lower uncertainty than evidence 2 from the visual perspective. Consequently, the unimodal-text decision strategy is selected for the final output. In this section, we provide a detailed explanation of the rules governing the strategy selection.

$$G_t^{m^l} = \sum_{j \in \mathcal{N}(i)} \alpha_{i,j} W_t^l m_j^{l-1} \tag{6}$$

$$G_v^{m^l} = \sum_{j \in \mathcal{N}(i)} \alpha_{i,j} W_v^l m_j^{l-1} \tag{7}$$

In the Eq. (7), $m^l$ indicates the node embedding update representation of the $l$th layer. $W^l$ indicates the weight parameter of the $l$th layer. $\alpha_{i,j}$ denotes the attention score between nodes $i$ and $j$ in the unimodal graph. The exact representation is given in Eqs. (8) and (9):

$$a_{i,j}^l = Softmax(e_{i,j}^l) \tag{8}$$

$$e_{i,j}^l = LeakyReLU(\vec{\alpha}[Wm_i \parallel Wm_j]) \tag{9}$$

where $\parallel$ is a splicing operation, and LeakyReLU represents Leaky Rectified Linear Unit.

To model inter-modal uncertainty, a heterogeneous graph structure $G_{mm}$ is incorporated following the reconstruction of the unimodal features. For $G_{mm}$, the GAT is continued to be used. Furthermore, to choose the appropriate decision strategy, the reconstructed features from the UEM are input into DFN. During the testing phase, this module refrains from updating its parameters, ensuring that decisions are based solely on the features of the current test sample, without being influenced by any external factors. This design makes it particularly well-suited for real-time fake news detection. However, the traditional D–S fusion approach [42] is not compatible with the MDF-FND framework; therefore, we propose a novel fusion method. Specifically, we first calculate the confidence score for each modality based on the normalized coefficient of variation [50], as defined in Eqs. (10) and (11). In the initialization of the D–S evidence theory, we map the confidence scores described above to the confusion level of each probability assignment function (also called BPA), which is used to represent the uncertainty estimate of each BPA for the final decision. The detailed calculation steps are presented in Algorithm 1. It is worth noting that through multiple experiments, we found that omitting the conflict factor $K$ leads to better results. Therefore, the influence of the conflict factor is omitted from the traditional D–S theory fusion in the decision strategy.

$$Score^T = sigmoid(\frac{\sqrt{\sigma_t}}{\mu_t} * scal) \tag{10}$$

$$Score^V = sigmoid(\frac{\sqrt{\sigma_v}}{\mu_v} * scal) \tag{11}$$

In the above equation, $\mu^t$ and $\mu^v$ denote the reconstructed mean of the text and image, respectively. $\sigma_t$ and $\sigma_v$ denote the reconstructed variance of the text and image, respectively. $scal$ is a scaling factor, and $sigmoid$ is an activation function that maps the input to [0,1].

In this manner, the dynamic weights of the two modalities are captured with respect to the final decision value. Subsequently, the confidence scores (normalized coefficients of variation) of each modality, along with the inherent uncertainty associated with each, are combined to derive the final decision opinion, as shown in Fig. 6. We fully leverage the advantage of D–S evidence-theoretic fusion. Specifically, when both modalities exhibit low confidence levels, the final classification must align with the low-confidence set; in this scenario, the strengths of the multimodal heterogeneous graph are utilized to determine the final decision value. Conversely, when both modalities demonstrate high confidence, the classification result is typically drawn from the high-confidence set, at which point the decision value is randomly selected from one of the modalities. If only one modality has high confidence, the final classification is determined by that modality. In cases of conflicting decisions between the two modalities, the final classification confidence is reduced, and the decision-making capabilities of the multimodal heterogeneous graph are adopted. To facilitate this, a threshold is established as a measurement criterion, detailed in Algorithm 1. The selection of this threshold is discussed further in Section 4.8.2.

**Algorithm 1:** D–S evidence theory algorithm for multimodal fake news detection

**Input:** Decision values from two unimodal perspectives $\mathcal{D}_t$ and $\mathcal{D}_v$ and their uncertainty scores $\mathcal{S}_t$ and $\mathcal{S}_v$
, respectively
**Output:** High-confidence decision-making strategies $\mathcal{P}$
Normalized decision values for unimodal text states:
$\mathcal{N}_t = \frac{1}{1+exp(-\mathcal{D}_t)}$, $\mathcal{N}_v = \frac{1}{1+exp(-\mathcal{D}_v)}$
Get two unimodal decision strategies for the object:
$m1[\mathcal{A}] = \mathcal{D}_{t_{out1}}$, $m1[\mathcal{B}] = \mathcal{D}_{t_{out2}}$
$m2[\mathcal{A}] = \mathcal{D}_{v_{out1}}$, $m2[\mathcal{B}] = \mathcal{D}_{v_{out2}}$
Fusion of evidence obtained from two modalities using uncertainty scores:
$m[\mathcal{A}] = m1[\mathcal{A}] * m2[\mathcal{A}] + m1[\mathcal{A}] * \mathcal{D}_v + \mathcal{D}_t * m2[\mathcal{A}]$
$m[\mathcal{B}] = m1[\mathcal{B}] * m2[\mathcal{B}] + m1[\mathcal{B}] * \mathcal{D}_v + \mathcal{D}_t * m2[\mathcal{B}]$
Get the uncertainty values of the two fused modes:
$m[\mathcal{A} \bigcup \mathcal{B}] = \mathcal{D}_t * \mathcal{D}_v$
Determine whether the overall uncertainty is greater than the threshold $\gamma$:
**if** $m[\mathcal{A} \bigcup \mathcal{B}] > \gamma$ **then**
  Decision values obtained using the theory of evidence are not credible:
    $uncertainty = True$
**end**
**else**
  Decision values obtained using the theory of evidence are credible:
    $uncertainty = False$
**end**
**if** $uncertainty = False$ *and* $\mathcal{D}_t > \mathcal{D}_v$ **then**
  The output of unimodal text should be used as a metric:
  $\mathcal{P} = max(m1[\mathcal{A}], m1[\mathcal{B}])$
**end**
**else if** $uncertainty = False$ *and* $\mathcal{D}_t < \mathcal{D}_v$ **then**
  The output of unimodal image should be used as a metric:
  $\mathcal{P} = max(m2[\mathcal{A}], m2[\mathcal{B}])$
**end**
**else**
  The output of the multimodal graph attention mechanism is used as a metric: $\mathcal{P} = \mathcal{G}_m$
**end**
**return** $\mathcal{P}$

### 3.4. Fake news detector

We follow previous research works [11,12,51], where we classify the robust features formed from each unimodal isomorphic graph $G_t^{m^l}$ and $G_v^{m^l}$ using a linear neural network with an activation function. The same operation is performed after the heterogeneous graph $G_{mm}$. Note that the two unimodal classifiers and the multimodal classifier are not used simultaneously; the final classification strategy is determined based on the confidence scores discussed in Section 3.3. Formally, this is expressed as

$$\hat{y}_n = Softmax(W_n h + b_n) \tag{12}$$

In the above equation, $W_n$ and $b_n$ are trainable parameters. To prevent the occurrence of a poor overfitting phenomenon, a dropout mechanism is added after the linear layer.

### 3.5. Regularized loss function

We attempted to optimize our classification model using the loss function defined in Eq. (13):

$$\mathcal{L}(\theta) = \frac{1}{N} Y_i log \hat{P}_i^0 + (1 - Y_i) log(1 - \hat{P}_i^1) \tag{13}$$

where $N$ indicates the total number of news samples.

However, using only $\mathcal{L}(\theta)$ as the final loss function leads to a variance collapse. As all sample point vectors converge to the optimum, the final distributional representation degenerates into a deterministic point representation, causing the model to lose its ability to learn multimodal uncertainty. To mitigate this negative phenomenon, a penalty term $\mathcal{L}_{reg}$ is introduced, which prevents the distribution's level of uncertainty from falling below the specified lower bound $\eta$. Consequently, an entropy-defined penalty term of the Gaussian distribution is introduced into the training framework, formally redefining the loss function as

$$\mathcal{L}_{pred} = \mathcal{L}(\theta) + \alpha \mathcal{L}_{reg} \tag{14}$$

where $\mathcal{L}(\theta)$ denotes the cross-entropy loss function defined in Eq. (15). $\mathcal{L}_{reg} = \mathcal{L}_{reg-text} + \mathcal{L}_{reg-image}$, Formally, $\mathcal{L}_{reg}$ can be expressed as

$$\mathcal{L}_{reg} = max(0, \eta - h(\mathcal{N}(\mu, \sigma^2))) \tag{15}$$

$\eta$ in the above equation as a lower bound to control the minimum level of uncertainty in the final distribution. In Section 4.8.3, the role of this hyperparameter is discussed in detail, including its influence on the overall effectiveness of the model. The multivariate Gaussian distribution is denoted by $h(\mathcal{N}(\mu, \sigma^2))$ and can be formally expressed as

$$h(\mathcal{N}(\mu, \Sigma)) = \frac{1}{2} log(det(2\pi e \Sigma)) \tag{16}$$

where $\Sigma$ indicates the covariance matrix. Because a diagonal array is used, the diagonal vector of $\Sigma$ is $\sigma$, and Eq. (16) can be transformed into Eq. (17):

$$\begin{aligned} h(\mathcal{N}(\mu, \Sigma)) &= \frac{1}{2} log(det(2\pi e \Sigma)) \\ &= \frac{2}{d}(log(2\pi) + 1) + \sum_{i=1}^{d} log\sigma_i \end{aligned} \tag{17}$$

where $d$ denotes the feature dimension.

For the final fake news detector, the Softmax function is used to convert logistic values into category probabilities. Let $\hat{P} = [\hat{P}_n^0, \hat{P}_n^1]$ be the final output probability vector and $Y_n$ be the ground-truth labeled value for $n$th news sample, where $\hat{P}_n^0$ and $\hat{P}_n^1$ denote the $n$th sample's true and false predicted probabilities, respectively. The MDF-FND framework minimizes the cross-entropy loss defined in Eq. (13) to train the model for enhanced detection of fake news in end-to-end manner. Previous studies have employed a penalty term with the standard normal distribution [48]. In subsection, we compare their performance against that of MDF-FND, which employ the entropy of the Gaussian distribution alone as the optimization function 4.7.3.

## 4. Experiments

In this section, we first present the public dataset used in the experiments and the basic setup of the experiments. Common baseline models used for fake news detection are also listed. In addition, the performance of the proposed dynamic fusion framework for fake news detection is experimentally evaluated. The superior performance of the MDF-FND framework is well demonstrated in relevant ablation experiments. Finally, we further explore the effective selection of each module and loss function by means of qualitative and quantitative analyses and analyze the impact of hyperparameters on the model.

### 4.1. Datasets

To demonstrate the effectiveness of the proposed model, we conducted experiments using three public datasets: Twitter, Weibo, and PolitiFact. These datasets originate from different language environments and provide valuable reference points for evaluating the generalization performance of the MDF-FND framework.

**Twitter dataset**. This dataset was collected and published by the MediaEval Benchmarking Initiative (Boididou et al. [23]) to evaluate the performance of multimodal models. The fake news in the datasets was sourced from news websites that were manually assessed to have low credibility, while the real news was crawled from well-known authoritative news sites such as the New York Times. Specifically, the dataset contains a large amount of social tweet content consisting of text and images, and its training set consists of 6000 fake news posts and 5000 real posts. The test set contains different types of breaking news, with up to 2000 posts. Posts containing only images or text are forcibly removed and not allowed to participate in the testing process of the model. Given its extensive data samples and diverse content types, this dataset closely resembles real news environments. Moreover, many posts in the Twitter dataset contain noise, which can significantly affect the final classification outcomes. Therefore, it is crucial to conduct experiments using uncertainty estimation and dynamic fusion strategies on this dataset.

**Weibo dataset**. To validate the robustness of the MDF-FND framework, the Weibo dataset is included, which belongs to a different linguistic environment than Twitter. This Chinese dataset has been widely used in recent studies to verify the superior performance of their fake news detectors. This dataset was first collected and publicly released by Jin et al. [4] and contains only Chinese posts and comment information. Weibo actively encourages users to report suspicious accounts, malicious hype, and comments related to trending events. The generation of fake posts involves two major steps: first, ordinary users report suspected fake news on the Weibo platform, but this alone cannot serve as the final judgment. Subsequently, a committee of well-known users conducts a further review to verify the authenticity of these reports. For non-rumor posts, those verified by the authoritative Chinese news agency Xinhua are used. Low-quality images and posts were kept in our experiments to further demonstrate the robustness of our classification model. The training set constitutes 70% of the entire dataset and the rest is divided into validation and test sets in a 1:2 ratio. In contrast to the Twitter dataset, the Weibo dataset features more general and universal content. Tweets often encompass trending topics and hashtags, while images frequently contain watermarked information. This type of content can introduce misleading signals that affect model decision-making during feature extraction. Furthermore, to assess the generalization ability of the MDF-FND framework across different languages, we included the Weibo dataset in our experiments.

**Table 2**
Statistics of three real-world multimodal news datasets.

| Dataset | Label | Number | Total |
|---|---|---|---|
| Twitter | Fake | 7021 | 12 995 |
| | Real | 5974 | |
| Weibo | Fake | 4749 | 9528 |
| | Real | 4779 | |
| PolitiFact | Fake | 367 | 801 |
| | Real | 434 | |

**PolitiFact dataset**. The PolitiFact dataset consists of information collected from the fact-checking website PolitiFact [27]. At PolitiFact, journalists and subject matter experts review political news and conduct fact-checking assessments to determine the authenticity of news articles. This dataset is well-balanced, primarily containing 434 pieces of real news and 367 pieces of fake news, most of which are related to U.S. politics. The PolitiFact dataset is introduced because it allows detection based on factual consistency checks, whereas the MDF-FND model is primarily designed for semantic detection. To explore the model's performance in the realm of fact-checking, we included this dataset in our comparative experiments.

Table 2 lists the statistical information of these three publicly-available datasets.

### 4.2. Experimental setup

**Parameter settings**. We adopt a common approach used in many previous studies [11,12,48]. The text embeddings are initialized using bert-base-chinese and bert-base-uncased sentence- and word-level annotations for the Weibo and Twitter datasets, respectively, with the obtained feature dimensions of $d^t$ equal to 768. To avoid the catastrophic forgetting of previous generalized knowledge [52], the parameters of the BERT model [46] were frozen during model training. For the image data, we standardized the image size to $224 \times 224 \times 3$ and fed it into the pre-trained ResNet-50 network [47], extracting features with a dimension of $d^v = 2048$. In the UEM module, which models intra-modal uncertainty, a linear network was employed to obtain the query, key, and value vectors for each modality. Prior to this, the layer normalization is applied. To prevent the risk of overfitting, dropout = 0.1 was considered for the attention linear layer, and each linear layer was followed by a nonlinear transformation using the GELU activation function, employing 12 heads. Experiments showed that a threshold $\gamma$ of 0.4 in Algorithm 1 performs excellently on the Twitter dataset. Similarly, for the Weibo dataset, the threshold $\gamma$ is set to 0.35. The reparameterized sample size was set to 5. The dropout of the GAT layer was set to 0.4, while the ELU activation function was employed. After we embedded the fake news detector into the GAT structure, it consisted of 2 linear layers and employed a dropout value of 0.4 and ReLU activation function for nonlinear mapping. The model was trained on a batch size of 128 over 40 epochs, with the learning rate initialized to 1e−4. For the PolitiFact dataset, the same setup as for the Twitter dataset is used.

**Evaluation Metrics**. We followed a previous study [4] and used the F1 score as the final assessment. Specifically, F1 score was calculated using the following formula:

$$F1 = \frac{2 \times P \times R}{P + R} \tag{18}$$

**Table 3**
Performance comparison of proposed MDF-FND with other baselines on the Twitter, Weibo, and PolitiFact datasets.

| Dataset | Method | ACC | Fake news | | | Real news | | |
|---|---|---|---|---|---|---|---|---|
| | | | P | R | F1 | P | R | F1 |
| Twitter | SVM-TS [54] | 0.529 | 0.488 | 0.497 | 0.496 | 0.565 | 0.556 | 0.561 |
| | GRU [55] | 0.634 | 0.581 | 0.812 | 0.677 | 0.758 | 0.502 | 0.604 |
| | CNN [56] | 0.549 | 0.508 | 0.597 | 0.549 | 0.598 | 0.509 | 0.550 |
| | BERT [46] | 0.706 | 0.648 | 0.540 | 0.589 | 0.715 | 0.636 | 0.673 |
| | SAFE [12] | 0.766 | 0.777 | 0.795 | 0.786 | 0.752 | 0.731 | 0.742 |
| | VQA [13] | 0.631 | 0.765 | 0.509 | 0.611 | 0.550 | 0.794 | 0.650 |
| | MVAE [11] | 0.745 | 0.801 | 0.719 | 0.758 | 0.689 | 0.777 | 0.730 |
| | EANN [9] | 0.648 | 0.810 | 0.498 | 0.617 | 0.584 | 0.759 | 0.660 |
| | SpotFake [10] | 0.892 | **0.902** | **0.964** | **0.932** | 0.847 | 0.656 | 0.739 |
| | BDANN [26] | 0.821 | 0.790 | 0.610 | 0.690 | 0.830 | 0.920 | 0.870 |
| | MCNN [29] | 0.823 | 0.858 | 0.801 | 0.828 | 0.787 | 0.848 | 0.816 |
| | MPFN [25] | 0.833 | 0.846 | 0.921 | 0.880 | 0.809 | 0.721 | 0.740 |
| | DIFF [35] | 0.783 | 0.810 | 0.803 | 0.806 | 0.758 | 0.786 | 0.772 |
| | CMMTN [24] | 0.903 | 0.870 | 0.917 | 0.892 | 0.927 | 0.881 | 0.903 |
| | COOLANT [31] | 0.900 | 0.879 | 0.922 | 0.900 | 0.923 | 0.880 | 0.901 |
| | MDF-FND | **0.947** | 0.876 | 0.962 | 0.917 | **0.983** | **0.941** | **0.961** |
| Weibo | SVM-TS [54] | 0.640 | 0.741 | 0.573 | 0.646 | 0.651 | 0.798 | 0.711 |
| | GRU [55] | 0.702 | 0.671 | 0.794 | 0.727 | 0.747 | 0.609 | 0.671 |
| | CNN [56] | 0.740 | 0.736 | 0.756 | 0.744 | 0.747 | 0.723 | 0.735 |
| | BERT [46] | 0.804 | 0.800 | 0.860 | 0.830 | 0.840 | 0.760 | 0.800 |
| | VQA [13] | 0.736 | 0.797 | 0.634 | 0.706 | 0.695 | 0.838 | 0.760 |
| | MVAE [11] | 0.824 | 0.854 | 0.769 | 0.809 | 0.802 | 0.875 | 0.837 |
| | EANN [9] | 0.794 | 0.790 | 0.820 | 0.800 | 0.800 | 0.770 | 0.780 |
| | BDANN [26] | 0.814 | 0.800 | 0.860 | 0.830 | 0.840 | 0.760 | 0.800 |
| | MPFN [25] | 0.870 | 0.887 | 0.849 | **0.868** | 0.855 | **0.891** | **0.873** |
| | MFIR [57] | **0.917** | **0.901** | 0.839 | **0.868** | **0.906** | 0.832 | 0.868 |
| | LIIMR [36] | 0.900 | 0.878 | 0.817 | 0.846 | 0.884 | 0.826 | 0.854 |
| | MDF-FND | 0.819 | 0.841 | **0.876** | 0.858 | 0.855 | **0.891** | **0.873** |
| PolitiFact | VGG-19[58] | 0.45 | 0.492 | 0.473 | 0.482 | 0.573 | 0.273 | 0.370 |
| | TextCNN [59] | 0.608 | 0.621 | 0.623 | 0.622 | 0.513 | 0.865 | 0.645 |
| | BERT [46] | 0.781 | 0.766 | 0.878 | 0.818 | 0.822 | 0.690 | 0.750 |
| | EANN [9] | 0.804 | 0.808 | 0.794 | 0.801 | 0.796 | 0.782 | 0.785 |
| | MVAE [11] | 0.812 | 0.803 | 0.835 | 0.819 | 0.866 | 0.747 | 0.802 |
| | SAFE [12] | **0.872** | **0.883** | 0.897 | **0.890** | **0.925** | 0.777 | **0.845** |
| | SpotFake [10] | 0.770 | 0.753 | 0.795 | 0.773 | 0.603 | 0.937 | 0.734 |
| | CAFE [32] | 0.848 | 0.847 | 0.860 | 0.853 | 0.762 | **0.948** | **0.845** |
| | MDF-FND | 0.795 | 0.712 | **0.955** | 0.816 | 0.690 | 0.875 | 0.772 |

where $P = \frac{TP}{TP+FP}$, and $R = \frac{TP}{TP+FN}$. The accuracy is formulated as follows:

$$ACC. = \frac{TP+TN}{TP+TN+FP+FN} \tag{19}$$

where $P$ denotes precision, $R$ denotes recall, $TP$ (True Positive) denotes the total number of posts predicted to be in the positive category, $FP$ (False Positive) refers to the total number of samples in the negative category that were predicted to have a positive label, $FN$ (False Negative) denotes the round-up of samples from the positive category that were predicted to be in the negative category, and $TN$ (True Negative) denotes the total number of samples that were predicted to be negative. The Adam optimizer [53] is used to optimize the model parameters to reach optimal decisions.

### 4.3. Baselines

Most previous research has focused on unimodal and multimodal fake news detection. To evaluate the superiority of our proposed model, we compared it with a baseline model that employs a single textual modality. In this comparison, we eliminated the dynamic decision-making component and directly input the noisy single-textual posts into the UEM module. A fully connected layer splicing was then used to output the predicted values.

(a) Fake news detection model based on single modality. The SVM-TS model [54] employs a machine learning approach to detect fake news through heuristic rules and linear multi-layer perceptron classifiers. To capture temporal feature information, the GRU model [55] implements a RNN for fake news detection. In addition, a CNN is employed to learn feature representations by framing related posts into fixed-length sequences, as introduced in [56]. Additionally, a model based on the pre-trained BERT [46] with a fully connected layer is constructed for detecting fake news from plain text input. Meanwhile, to assess the performance of the unimodal image baseline model, VGG-19 [58] and TextCNN [59] are incorporated into the comparative experiments.

(b) Fake news detection model based on multimodal features. VQA [13] aims to answer questions about the given images. The original VQA model is designed for multi-class classification tasks. SAFE [12] is a similarity-aware cross-modal fusion function for multimodal FND. The MVAE model [11] extracts modal information from text and images and uses a multimodal variant encoder for simple fusion. [9] constructed an event adversarial neural network (EANN) that uses an event learner to learn the feature representations of text and images in an article; however, additional auxiliary features will increase the cost of detection. Intuitively, the SpotFake model [10] can use large-scale textual and visual pre-trained models to extract features for each modality and concatenate them into a multimodal representation of the news, which is then placed into a binary classifier for detecting fake news. MCNN [29] considers the consistency of different modalities and captures the global characteristics of social media information. BDANN [26] is a BERT-based domain adaptive neural network for detecting multimodal fake news. LIIMR [36] utilizes both intra- and inter-modal relationships for fake news detection. MFIR [57] recognizes that traditional fusion methods, such as union or addition, can enhance the performance of multimodal fake news detection to some extent. However, this approach is mainly limited by shallow cross-modal feature fusion. Therefore, this model focuses on identifying inconsistent semantics across modalities to achieve interpretable fake news detection. MPFN [25] is a progressive fusion network designed for multimodal fake news detection, capable of capturing representation information for each modality at different levels. It uses mixers to facilitate fusion between modalities at the same and different levels, thereby establishing strong connections between them. DIIF [35] employs a dynamic internal and cross-modal fusion model for multimodal fake news detection, dynamically learning the interactions and fusions among different modalities. The SAFE [12] model detects fake news on social media using a similarity-aware approach. CAFE [32] enhances the accuracy of fake news detection by intuitively and adaptively aggregating unimodal features and cross-modal correlations. CMMTN [24] enhances the performance of fake news detection by jointly modeling both the inter- and intra-modal relationships of multimodal information, while masking irrelevant context. COOLANT [31] is a cross-modal contrastive learning framework for multimodal fake news detection, aiming to achieve more accurate image–text alignment. It implements an attention mechanism with an attention-guided module to help effectively and interpretably aggregate the aligned unimodal representations and cross-modal correlations.

### 4.4. Main results

This subsection compares the performance of our proposed MDF-FND with that of existing baselines and provides specific analytical results.

The experimental results in Table 3 illustrate the superiority of the proposed framework on the three publicly available datasets. Specifically, we derived the following results:

**Table 4**
Comparison of trainable parameters and computational speed. FLOPs: amount of floating point arithmetics.

| | MDF-FND | EANN | SpotFake | CAFE |
|---|---|---|---|---|
| Params | 30.9M | 143.70M | 124.37M | 0.68M |
| FLOPs | 18.82G | 19.63G | 30.42G | 0.01G |

(a) Among the unimodal methods, especially on textual modalities, the PLMs represented by BERT demonstrate irreplaceable capturing ability. On the Twitter dataset, it outperforms the traditional support vector machine (SVM) by almost 9.3%. This reflects the effectiveness of employing textual unimodality for fake news detection, as visual fakery is usually not easily captured.
(b) Multimodal fake news detection methods (e.g., MVAE [11] and EANN [9]) usually have more sensitive insights than unimodal text detection methods because they usually take into account the inconsistent semantic information of both modalities, text and image, in their models. However, these methods fail to address the incidental uncertainty arising from the noise often found in multimodal news.

On two public datasets, Twitter and Weibo, our proposed MDF-FND consistently outperforms the latest baseline methods. The **4%** improvement in accuracy over the optimal baseline model was achieved. Correspondingly, the F1 Score increases by **6%**. In this paper, the data uncertainty problem caused by the noise in multimodal posts circulating in social networks is elaborated, while the uncertainty modeling mechanism is introduced in our model, which is able to capture the unimodal features in complex heterogeneous multimodal post data. In addition, the heterogeneous GAT shifts the goal to modeling the inter-modal uncertainty, and the mapped features have stronger robustness; thus the performance is effectively improved. However, because the two datasets originate from distinct linguistic environments, tweets on the Twitter platform are typically brief, while Weibo posts often include trending topic tags and a broader range of content expressions. Consequently, there are substantial differences between the results.

The MDF-FND framework exhibits poor performance on the PolitiFact dataset, and our analysis reveals two primary reasons for this. First, the PolitiFact dataset relies on factual verification, where external knowledge is crucial for guiding detection. Thus, relying solely on semantic information is inadequate to accurately identify fake news. Second, the dataset contains numerous posts without accompanying images, and the MDF-FND framework struggles to process these posts effectively. If a multimodal post lacks information from one modality, the dynamic fusion approach is divided into traditional unimodal fake detection tasks, which presents a considerable challenge for the model. In the following experiments, the overall performance of the MDF-FND framework is mainly evaluated on the two public datasets, Twitter and Weibo.

In summary, the advantage of the MDF-FND framework lies in its ability to model both intra-modal and inter-modal uncertainty by utilizing the strengths of the attention mechanism, thus improving the robustness of the detection model. However, its limitation is that, in the context of fact-checking news, it cannot incorporate additional considerations, particularly clues relevant to fact verification. This represents a promising direction for future research.

### 4.5. Computational complexity

The computational complexity of the MDF-FND framework is a crucial aspect to address, particularly in the context of large-scale datasets and real-time applications. In this section, we will analyze the computational complexity associated with the various components of the method we proposed.

In the feature extraction phase, the frozen parameters of the pre-trained bert-base-uncased and ResNet-50 are employed to extract semantic information from text and images, respectively. The computational complexity of BERT encoding and attention calculation for news posts is $O(n^2 \cdot d)$, where $n$ denotes the number of tokens in each news, and $d$ represents the embedding dimension. For image feature extraction, the key factor is the number of $ROIs$(Regions of Interest). Let the number of ROIs extracted from the image be denoted as $l$, with each $ROI$ feature represented as $e_i^v \in \mathbb{R}^{d_v}$, where $d_v$ is the dimension of each $ROI$. Assume that the feature extraction time for each $ROI$ is at a constant level, $O(1)$, the total extraction complexity is $O(l)$. The complexity of the convolution operation is expressed as $O(H \cdot W \cdot C \cdot K^2)$, where $H$ and $W$ are the height and width of the input image, $C$ is the number of input channels, and $K$ is the size of the convolution kernel. The computational complexity of the average pooling operation is $O(C \cdot L)$, where $C$ denotes the number of channels and $L$ denotes the length of the input sequence.

In the uncertainty modeling phase, the multi-head self-attention module is primarily employed for operations. Assume there are $k$ attention heads, and the input representation of each attention head is $H^{(i)} \in \mathbb{R}^{T \times D/2k}$, where the total complexity is $O(k \cdot T^2 \cdot D)$, with $T$ representing the length of the feature sequence and $D$ the size of the hidden layer. The complexity analysis for the two unimodal graph structures and one multimodal graph structure includes both the attention weight calculation and the update of node features. The overall complexity is $O(E \cdot F)$, where $E$ is the number of edges in the graph, which is determined by the length of the feature vectors of the two modalities. $F$ represents the dimensionality of the features of nodes.

Since sampling is non-differentiable, it affects the gradient propagation of the training process. However, the complexity of sampling from a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ is the constant level, marked $O(1)$, therefore does not significantly increase the overall complexity. The same is true for the complexity of the evidence fusion module.

The final classification module uses MLP to discriminate the two-dimensional robust features with a complexity of $O(H \times W \times L)$, where $H$ is the number of hidden units in the MLP hidden layer, $W$ is the width of the input feature vector, and $L$ is the length of the input sequence.

It costs around 10 h to train MDF-FND for 40 epochs on a single NVIDIA A40 GPU. The computational complexity analysis in Table 4 indicates that MDF-FND requires fewer trainable parameters and achieves significantly faster computational speed compared to EANN and SpotFake. CAFE requires amazingly low computational complexity by using simple auto-encoders and MLPs, but the performance is not good enough. In conclusion, the MDF-FND framework is well-suited for fake news detection tasks that require real-time processing.

### 4.6. Ablation studies

To verify the effectiveness of the proposed fusion strategy, the ablation analyses are conducted to evaluate the impact of each component on the performance of the MDF-FND framework. In each experiment, a specific component, either the UEM or DFN, is systematically removed and the model is retrained. The following variants of MDF-FND are implemented:

1. MDF-FND w/o UEM-Text. The uncertainty module used to model unimodal text is removed. The UEM module features are fed with captured visual uncertainty information and features extracted after BERT [46] into the dynamic fusion module, and Gaussian modeling for text features is not removed to better represent the modeling capabilities of DFN.

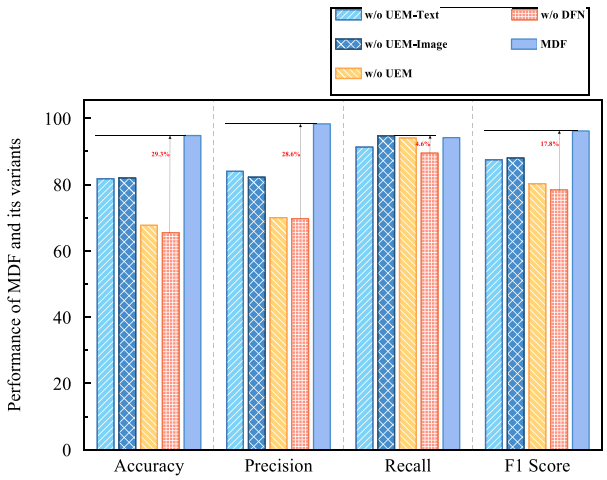



**Fig. 7.** Visualization of the comparative performances of MDF-FND and its variants.

2. MDF-FND w/o UEM-Image. The uncertainty module for modeling unimodal images is removed. The UEM module features with captured textual uncertainty information and the extracted features from the pre-trained ResNet-50 network are fed into the dynamic fusion module. Similarly, Gaussian modeling for visual features is not removed.
3. MDF-FND w/o UEM. The uncertainty module used to model unimodal text and images is eliminated. We assume that the two unimodal modalities, text and image, have the same confidence score and feed directly into the dynamic fusion module using BERT [46], with a simple splicing of text and image features extracted by the pre-trained ResNet-50 network.
4. MDF-FND w/o DFN. The dynamic fusion module is removed. The decision-making capability of the model relies only on modeling robust intra-modal uncertainty features extracted from unimodal text and unimodal images by simple splicing operations.

The results of our experiments on the Twitter dataset are visualized in Fig. 7. The overall performances of the uncertainty modeling modules excluding either unimodal text or unimodal images, that is, w/o UEM-Text and w/o UEM-Image, both degraded. Specifically, when excluding the uncertainty modeling module for unimodal text, the accuracy shows a significant decrease. This exemplifies the extent to which unimodal text affects the accuracy of fake news detection and further illustrates the importance of uncertainty modeling. Similarly, when the uncertainty module used to model unimodal images was removed, the overall model performance also decreased. However, the extent of the effect of the above two methods compared to that after the direct deletion of the UEM module, i.e., w/o UEM, is smaller. When the UEM module was directly deleted, the overall performance exhibited a sharp decline. The w/o DFN method, which relies only on simple splicing instead of the dynamic fusion strategy, shows high degrees of degradation in both accuracy and the F1 score.

It is clear that excluding the DFN in our proposed dynamic fusion architecture realizes poor results, with the worst accuracy directly decreasing by up to **29%**, further illustrating the superior performance of our proposed dynamic fusion architecture. This is attributed to the superiority of the dynamic fusion framework, which is capable of adjusting the final decision-making strategy based on the dynamic weights held by each modality using the normalized coefficient of variation as the fusion weight. The performance of the w/o UEM variant used to indicate the deletion of uncertainty modeling was also significantly degraded, which is a good indication of the adaptability of uncertainty modeling to our dynamic fusion framework.

**Table 5**
Comparison of optimal results on Twitter dataset between MDF-FND framework using cross-attention and using GAT mechanism.

| Model | | Accuracy | Micro precision | Micro recall | Micro F1-Score |
|---|---|---|---|---|---|
| MDF-FND-GAT | max | 94.791 | 98.313 | 94.159 | 96.191 |
| | min | 59.924 | 68.186 | 79.908 | 73.583 |
| MDF-FND-Att | max | 85.147 | 87.668 | 91.596 | 89.589 |
| | min | 63.986 | 69.100 | 87.515 | 77.224 |

### 4.7. Qualitative analysis

In this subsection, numerous experiments are conducted to prove the validity of the model. The first one combined the traditional cross-attention mechanism with our DFN module to form an MDF-FND framework using the cross-attention mechanism; however, this did not work as well as expected. This experiment is mainly used to show that our proposed UEM module is a perfect fit with the DFN module and that the role of uncertainty modeling in MDF-FND cannot be replaced by the traditional simple attention mechanism. In addition, the DFN module was replaced with many existing methods. However, the results still did not show excellent performance.

#### 4.7.1. MDF-FND using cross-attention mechanism vs. MDF-FND using graph-attention mechanism

We also continued the approach of employing cross-attention in the MDF-FND framework, as proposed in numerous previous studies, but obtained poor experimental results, as shown in Fig. 8(a).

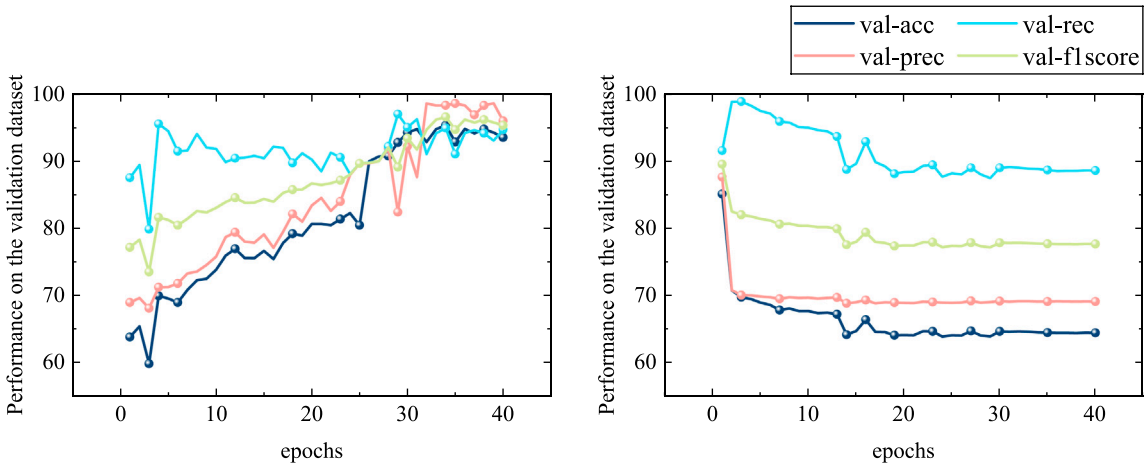
As can be seen in Fig. 8(a), the value of F1 score constantly shows a stepwise decrease during the model testing phase, while accuracy and precision show different degrees of decrease. This indicates that the traditional cross-attention mechanism does not fit well with our proposed dynamic fusion framework. By contrast, using the GAT mechanism can substantially improve the effectiveness of the model, the effect of which can be seen in 8(b). Although there is an unanticipated decrease in efficiency in a small region, the overall trend of the model is positive, and the best F1 score is improved by **7%** compared to the effect of using cross-attention. Table 5 presents the best and worst performance capabilities of the MDF-FND framework with cross-attention versus the MDF-FND framework with the GAT mechanism on the Twitter dataset. This is attributed to the fact that the GAT mechanism can better capture robust feature representations obtained using unimodal uncertainty modeling, whereas the cross-attention-only mechanism does not adequately represent the heterogeneous properties of features and is not able to model inter-modal uncertainty.

#### 4.7.2. Effectiveness of the dynamic fusion framework

To illustrate the uncertainty modeling ability of the proposed dynamic fusion framework for the robust representation features of single modalities, the dynamic fusion module DFN was eliminated and replaced with the following methods:

- **MDF-FND-Concat** indicates a simple connection using multimodal features.
- **MDF-FND-FC** uses a fully connected layer to fuse multimodal features.
- **MDF-FND-Att** fuses multimodal features and attention mechanisms.
- **MDF-FND-CoA** fuses multimodal feature information and common attention mechanisms.

Fig. 9 shows the performance of MDF-FND with different fusion strategies. From the experimental results, it can be observed that the method using the DFN module as the fusion strategy for MDF-FND combined with uncertainty modeling outperforms the other fusion modules on both datasets, which demonstrates the superior performance of our model. The performance of MDF-FND under the MDF-FND-FC and




(a) MDF-FND with cross-attention. (b) MDF-FND with GAT.

**Fig. 8.** Here, val-acc refers to the accuracy of each model on the test set, val-prec refers to the accuracy of each model on the test dataset, val-rec refers to the recall of each model on the test dataset, and val-f1score refers to the F1 score of each model on the test dataset. Note that the above experimental results all are for the Twitter dataset.

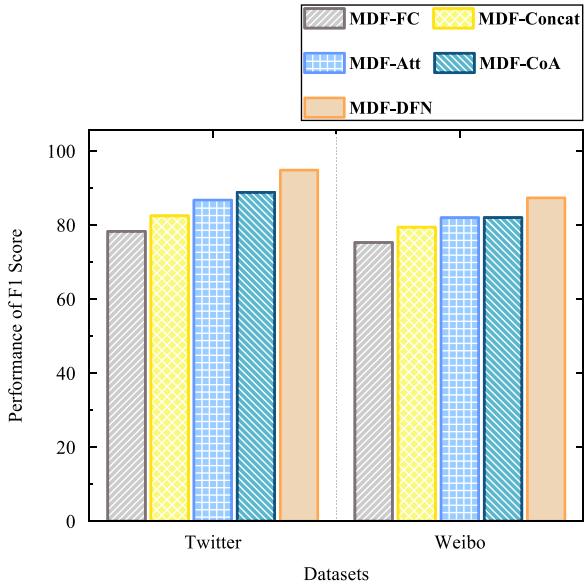



**Fig. 9.** Performance of the MDF-FND framework with different fusion strategies on two public datasets.

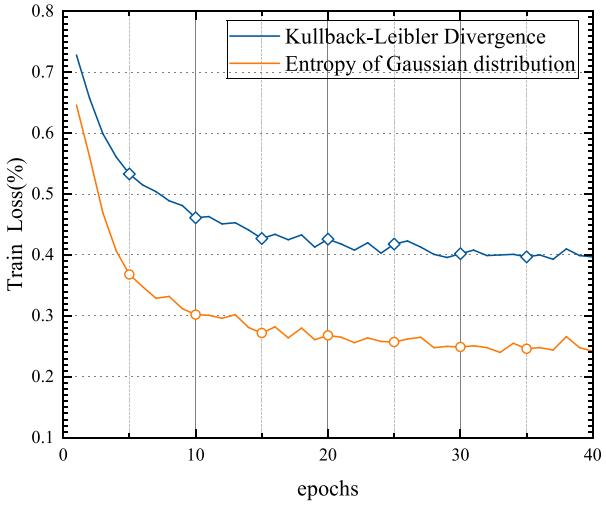



**Fig. 10.** Performance of the MDF-FND framework with different fusion strategies on two public datasets.

MDF-FND-Concat mechanisms is slightly inferior to that of MDF-FND-Att and MDF-FND-CoA using the attention mechanism, which also fully illustrates the strong ability of the attention mechanism to capture complementary feature information. Meanwhile, the fully connected layers and simple splicing operations are also limited by the fact that they ignore the complex logical relationships between text and visual modalities.

#### 4.7.3. Comparison of loss functions

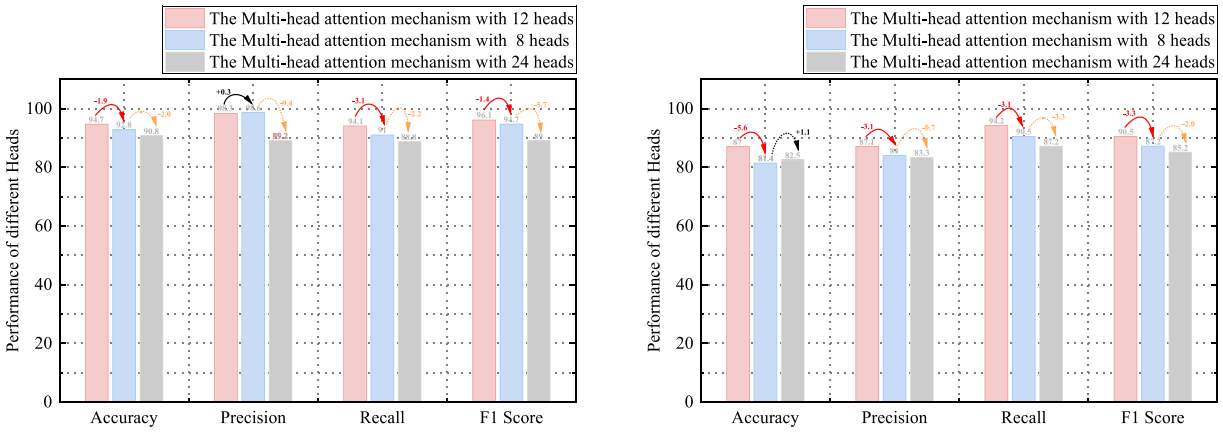
The Kullback–Leibler scatter (K–L) scatter is a measure of the difference between two probability distributions. In uncertainty modeling, the K–L scatter is commonly used to compare the level of differentiation between a modeled probability distribution and standard normal distribution, which is attributed to the fact that data uncertainty tends to be modeled in the form of a normal distribution. Formally, the penalty term using K–L scatter can be defined as

$$\begin{aligned}\mathcal{L}_{kl} &= KL(\mathcal{N}(x;\mu,\sigma^2) \parallel \mathcal{N}(\epsilon;0,I)) \\ &= -\frac{1}{2}(1+log(\sigma^2)-(\mu)^2-(\sigma)^2)\end{aligned} \tag{20}$$

where $KL(\cdot \parallel \cdot)$ indicates the Kullback–Leibler divergence between the two probability distributions.

A small K–L scatter value indicates that the degree of differentiation between the model distribution and standard normal distribution is small, and the model fits the distribution of the real data well. Therefore, the model's ability to model uncertainty can be continuously improved by optimizing the model to reduce the K–L scatter. K–L scatter is also adopted as the optimization function of the MDF-FND framework. To achieve this, the optimization algorithm of MDF-FND is redefined according to Eq. (21). A comparison of entropy, which is a metric that describes the uncertainty of a random variable of a certain distribution, using a Gaussian distribution is shown in Fig. 10.

$$\mathcal{L}_{pred} = \mathcal{L}(\theta) + \alpha\mathcal{L}_{kl} \tag{21}$$




(a) Twitter dataset.

(b) Weibo dataset.

**Fig. 11.** Comparison on Twitter and Weibo datasets using multi-head attention mechanism with 8, 12, and 24 heads.

where $\mathcal{L}(\theta)$ is the cross-entropy loss function for the fake news detection task defined by Eq. (13), and $\alpha$ is the penalty term coefficient.

As shown in Fig. 10, the training loss of the MDF-FND framework using K–L scatter is larger than that of the MDF-FND framework using the entropy of the Gaussian distribution as the penalty term, especially in the later stages of model training, where it fails to converge to a reasonable interval over time. In addition, comparing the convergence speeds, the framework employing K–L scatter is relatively slow compared to the framework employing the entropy of the Gaussian distribution as the loss. Specifically, the fusion framework using the entropy of the Gaussian distribution as the penalty term gradually converged into the optimal interval in the 15th epoch; however, this was relatively difficult for the model using K–L scatter. Meanwhile, in the late stage of model training, the MDF-FND framework utilizing the entropy of the Gaussian distribution can predict the fluctuation of the data in the unknown interval well, while the framework using K–L scatter cannot. In summary, both in terms of convergence speed and performance capability, using the entropy of the Gaussian distribution as the final penalty term is much better than using K–L scatter. This is attributed to the entropy of a Gaussian distribution, which encourages the model to generate a broad distribution rather than one resembling the standard normal distribution. This is beneficial in Bayesian optimization because the model is forced to quantify the uncertainty of the prediction. Based on the above conclusions, the entropy of the Gaussian distribution was incorporated as a penalty term into the loss function of the MDF-FND framework to encourage the model to learn a broader distribution.

#### 4.7.4. Research on sustainability

Different regions, cultural backgrounds, and linguistic environments result in diverse understandings and manifestations of fake news. Research on the sustainability of fake news detection frameworks will foster fairness in global fake news detection efforts. In this subsection, the contributions of the sustainability of the MDF-FND framework are clearly outlined.

Specifically, it includes two main aspects. First, the MDF-FND framework plays a significant role in **technological evolution**. Traditional methods like concatenation or weighted fusion often fail to dynamically adjust the contributions of each modality based on the actual data needs. The MDF-FND framework overcomes this limitation by introducing a dynamic adjustment mechanism. This development will encourage other multimodal fake news detection models to become more flexible and adaptable to a broader range of news types. Furthermore, it will generate positive **social impacts**. By improving computational efficiency and reducing resource consumption, the application of the framework to real-world scenarios will assist fact-checkers in mitigating the impact of fake news. This will help maintain social fairness and harmony, promote the healthy development of the information environment, and strengthen societal cohesion and unity.

### 4.8. Quantitative analysis

In this subsection, we analyze the role of the relevant hyperparameters involved in the model and their impact on the final effect of the MDF-FND framework in detail.

#### 4.8.1. Parameter analysis of the attention mechanism in the UEM module

The UEM is primarily responsible for modeling uncertainty within each modality, in which we mainly use the multi-head attention mechanism. To evaluate the impact of the number of heads in the multi-head attention mechanism on overall performance, comparative experiments were conducted on the Twitter and Weibo datasets, with the results shown in Fig. 11. The results show that on both the Twitter and Weibo public datasets, excellent performance can be achieved when the number of heads is set to 8. However, as the number of heads decreases, the overall performance also show a small decrease.

In Fig. 11(a), employing a multi-head attention mechanism with 12 heads on the Twitter dataset is identified as a promising strategy for modeling uncertainty, although it has a performance degradation of 0.3 in terms of precision compared to the multi-head attention mechanism with 8 heads. Meanwhile, the effect on the Weibo data is shown in Fig. 11(b), where the 12-head attention is superior to 8-head attention, both in terms of accuracy and F1 score. However, it is not the case that more heads are more effective, and we conducted further experiments that showed a drop in performance after setting the number of attention heads to 24. This is mainly because when the number of heads is set to 12, the model's performance has already saturated, and when 24 heads are used, redundant representations appear due to the excessive number of heads. This result is consistent with that of [60]. For the subtask of modeling the uncertainty within each modality, excessive subspaces are not necessary to capture complex features and achieve excellent performance.

#### 4.8.2. Parameter analysis in the DFN module

As the most important part of the dynamic fusion framework, the DFN module, which adopts the graph attention mechanism and D–S evidence theory, directly affects the final dynamic decision. To better balance the contributions of the two modalities, the normalized

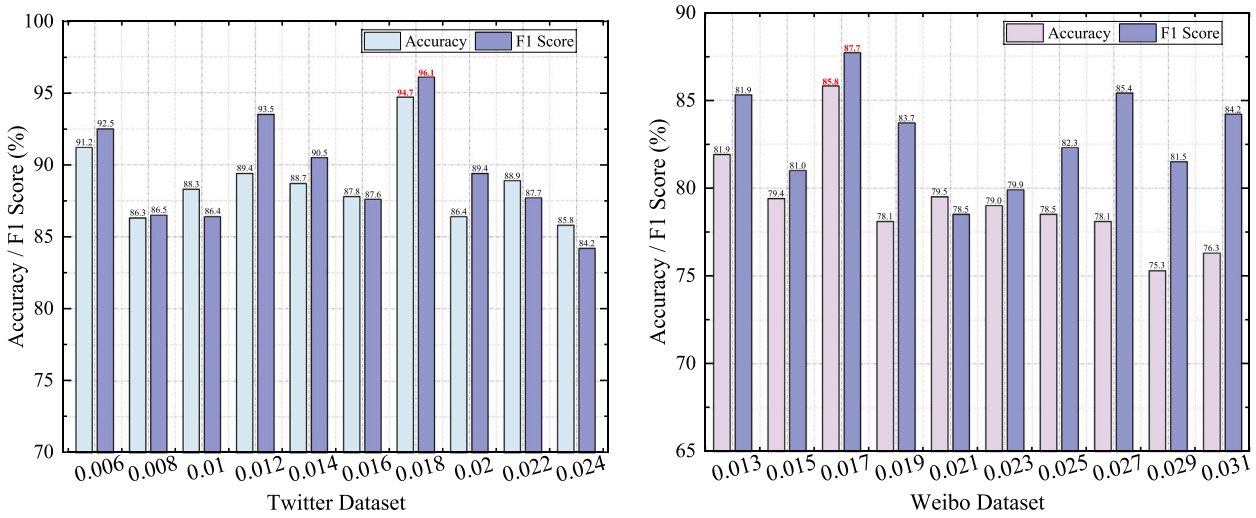



**Fig. 12.** Using the grid search method to identify the optimal values of the hyperparameter $\eta$ for both the Twitter and Weibo datasets.

**Table 6**
Impact of the threshold $\gamma$ used for dynamic fusion in the DFN module on the overall effect on publicly available Twitter and Weibo datasets.

| Dataset | $\gamma$ (Threshold) | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|---|
| Twitter | 0.25 | 0.76 | 0.79 | 0.90 | 0.84 |
| | 0.50 | 0.94 | 0.98 | 0.94 | 0.96 |
| | 0.75 | 0.92 | 0.98 | 0.91 | 0.94 |
| Weibo | 0.25 | 0.76 | 0.79 | 0.90 | 0.84 |
| | 0.50 | 0.81 | 0.85 | 0.89 | 0.87 |
| | 0.75 | 0.72 | 0.73 | 0.94 | 0.82 |

coefficient of variation of each modality is considered an important measure and compared with the hyperparameter in the DFN, which also serves as $\gamma$ of the confidence threshold. Therefore, this hyperparameter also directly influences the final dynamic fusion strategy. Several sets of experiments were conducted on the Twitter dataset to verify the effect of this parameter on the final classification results. The experimental results are presented in Table 6.

The DFN module plays a pivotal role in the MDF-FND framework. As the key hyperparameter $\gamma$ is used for dynamic fusion by the DFN module, it mainly assumes the role of selecting the final strategy. Relying on the strong adaptation capability of the heterogeneous graph structure, when a certain modality is filled with a large amount of noise and data uncertainty occurs, the DFN module decides whether to classify it with unimodal uncertainty modeling capability through the threshold $\gamma$. As shown in Table 6, the value of $\gamma$ should be selected to fit a specific distribution satisfied by the dataset, and numerous randomized experiments should be conducted for specific datasets to finalize the threshold. Six sets of experiments were conducted with $\gamma$ values of 0.25, 0.5, and 0.75, each transformed at 50% and 30%. The experimental results show that the effect of $\gamma$ values on the Weibo dataset is small relative to that on the Twitter dataset, with the maximum difference in F1 reaching **0.05%**. However, for the Twitter dataset, there were some differences in the selection of $\gamma$ values, with the best performance capability being **0.10%** more than the worst.

#### 4.8.3. Parameters in the optimization algorithm

The cost function is responsible for evaluating the variance between the MDF-FND framework and the real results, and therefore also its performance. Using only the cross-entropy loss function will lead to the collapse of the modeled variance and degradation of the distribution representation from the original high-dimensional Gaussian distribution to a deterministic point embedding approach, which further leads to a reduction in the model's generalization ability for multimodal information. Therefore, the entropy of the Gaussian distribution is added as a penalty term to reduce the impact of modal uncertainty on the results. The hyperparameter $\eta$, which acts as the lower bound for the model's minimum uncertainty output, has a direct impact on the final decision-making performance. To identify the optimal value for the MDF-FND framework, we employed a grid search strategy using the Optuna library [61], with a search space of $Float\{0.001, 0.05\}$. We set the number of trials to 10 and step size to 0.002, and we report the optimal convergence range on the Twitter and Weibo datasets in Fig. 12. Consequently, the best values for $\eta$ are 0.018 for the Twitter dataset and 0.017 for the Weibo dataset. The experimental results further indicate that the MDF-FND framework is highly sensitive to the choice of $\eta$.

### 4.9. Case study

To investigate the effectiveness of the dynamic fusion mechanism in multimodal fake news detection more deeply, we conducted a set of case studies, as shown in Fig. 13. Four representative cases from the Twitter and Weibo datasets are exemplified, presenting the overall uncertainty scores of the two modalities in detail, along with their respective unimodal uncertainty scores. The uncertainty score reflects the credibility of the modality for the final prediction outcome and is inversely related to uncertainty. As shown in Algorithm 1, when the value is high, particularly when it is larger than the set threshold, the decision opinion of the multimodal heterogeneous graph is directly used. When this value is low, we focus on the decision results for unimodal text and unimodal images. In the first case, where the overall uncertainty is relatively low and there is a large difference between the uncertainties of the two unimodal modalities, it is clearly observable that the unimodal text has a lower uncertainty score. Therefore, the MDF-FND framework adopts the decision opinion of the image. The opposite situation occurs in the second case, where the confidence level of the unimodal text is much higher than that of the unimodal image. Therefore, final decision power is granted to the unimodal text. When the overall uncertainty score is high, the decision-making power of the heterogeneous GNN is considered, as shown in case four. Obtaining a plausible decision from only two unimodal uncertainty scores may produce poor results. For example, in the third case, the unimodal image has a much larger uncertainty score than the text, but the image itself is of low quality, so the data tend to have uncertainty. Fortunately, when the overall uncertainty is considered, the final decision no longer relies on a low-quality unimodal image with a high uncertainty score but rather on the decision value of a multimodal heterogeneous image, which avoids catastrophic results to some extent.

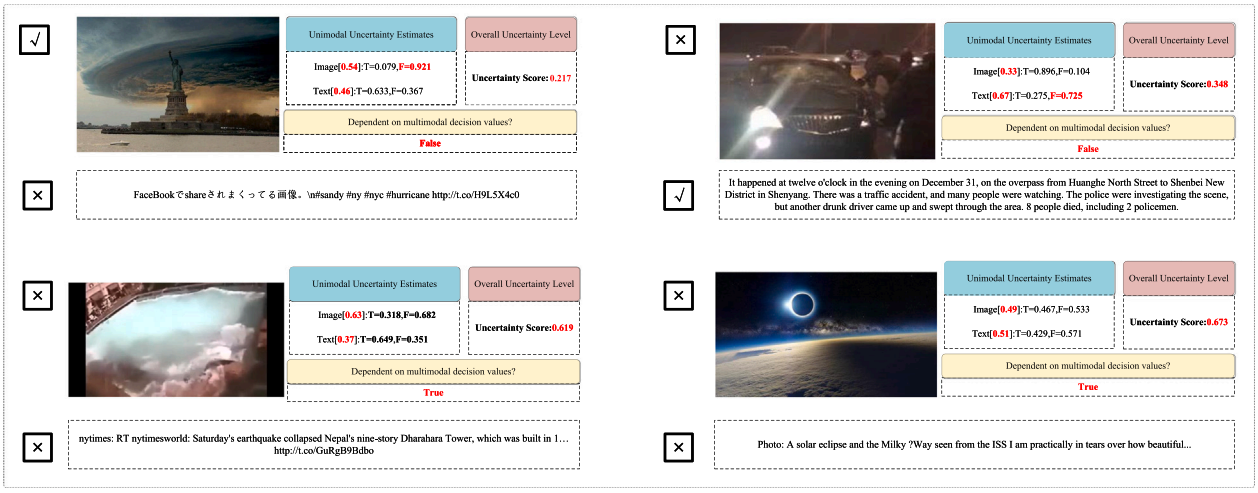



**Fig. 13.** Case study examples for the dynamic fusion framework. $T$ and F figure represent the predictions of true news and false news, respectively. The * in unimodal[*] represents the uncertainty score in the modality.

## 5. Limitations and potential research directions

Our framework named MDF-FND mainly has two limitations. On the one hand, it is evident that the computational complexity of the self-attention mechanism is $O(k \cdot N^2 \cdot d)$. As the sequence length increases, the effect of the number of heads $k$ becomes negligible, so it can be ignored. Therefore, the complexity can be expressed as $O(N^2 \cdot d)$, which is clearly impractical when processing long sequences. On the other hand, although the proposed framework demonstrates excellent performance in modeling uncertainty, it is limited to data processing within a single modality. During feature fusion, there are often multi-modal samples that are weakly aligned or misaligned, which are referred to as cross-modal noise [62]. Unlike modality-specific noise, cross-modal noise resides in a higher-level semantic space.

To address the above challenges, our future potential research directions lies in the following aspects. First, we will employ a linear attention mechanism, such as the Polarity-aware Linear Attention in PolaFormer [63], to facilitate uncertainty modeling in multimodal fake news detection, making it more suitable for widespread deployment. Additionally, we will adopt a cross-modal denoising approach to address redundant information representations generated during multimodal interactions.

## 6. Conclusion

Multimodal user posts containing fake news on social media involve data uncertainty caused by unimodal noise. Therefore, in this paper, a dynamic fusion framework called MDF-FND was proposed that can model inter- and intra-modal uncertainty while providing final decision results using a dynamic fusion strategy. The MDF-FND framework first contains a UEM module for performing uncertainty modeling, which maps the unimodal text and unimodal image features through multi-head attention mechanism mapping to a latent subspace that satisfies a Gaussian distribution. Subsequently, a DFN module employing the GAT mechanism with evidence theory is employed to conveniently capture inter-modal uncertainty information and is responsible for dynamically balancing the weights of the two modalities. Multiple sets of experiments on three publicly available datasets, Twitter, Weibo, and PolitiFact, demonstrated the effectiveness of the proposed framework.

Recent advances in LLMs have demonstrated excellent performance in various NLP endeavors. However, they still face significant challenges in multimodal fake news detection tasks. For example, it is mentioned in [63] that although complex LLMs like GPT3.5 can provide desirable multi-perspective justifications for exposing fake news, they are still not as effective as basic SLMs, such as fine-tuned BERT.

## Ethical consideration

In this section, we review our paper for the following ethical considerations:

- **Avoid using Generative AI and AI-assisted technologies during the writing process**. We hereby declare that we have never used any form of generative artificial intelligence technique or AI-assisted technology in the writing of this manuscript to ensure that our content maintains the highest degree of originality, accuracy, and ethical responsibility.
- **Avoid harm**. This manuscript focuses on the design of a deep learning framework for multimodal fake news detection in which participants were not harmed in any way.
- **Transparency**. We have provided the source code of the design framework and a link to the repository of the data used in the study, which we will upload as soon as the paper is accepted. To ensure sufficient transparency of our work, we have provided detailed experimental parameters as well as evaluation metrics in Section 4.2 and analyzed the impact of the involved parameters on the final performance in Section 4.8.
- **Public Good**. Our proposed MDF-FND framework is only used to output information about the truthfulness of a certain multimodal news sample, which does not have any negative impact on social opinion.
- **Confidentiality and Data Security**. We used three publicly available datasets, Twitter, Weibo, and PolitiFact, to evaluate our model, which have been used in many previous studies and therefore do not have any data security issues or confidentiality concerns.

In summary, we have determined that our manuscript does not present any potential ethical risks in terms of harm, public good, transparency, confidentiality, or data security.

## CRediT authorship contribution statement

**Hongzhen Lv:** Writing – review & editing, Writing – original draft, Validation, Software, Conceptualization. **Wenzhong Yang:** Supervision, Funding acquisition, Conceptualization. **Yabo Yin:** Supervision, Conceptualization. **Fuyuan Wei:** Formal analysis, Data curation. **Jiaren Peng:** Formal analysis, Data curation. **Haokun Geng:** Software.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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